

AI-Powered Data Quality & Validation Framework

Project understanding and Research

Project Description

This project builds an AI-assisted data quality and anomaly detection framework, applied to retail/e-commerce transaction data. The system validates incoming transaction datasets for structural correctness, detects anomalies and data entry errors, scores each dataset's overall quality, and generates human-readable reports summarizing the findings. The framework combines rule-based validation with AI-assisted analysis and machine learning-based anomaly detection.

Use Case

Retailers and e-commerce platforms process high volumes of transactions daily and need to catch data quality issues before they contaminate downstream reporting and analytics. This framework serves two connected purposes:

- **Data Quality Assurance:** Ensuring transaction datasets are structurally sound (no missing/invalid fields, values within expected ranges) before any downstream analysis or ML modeling is performed on them.
- **Anomaly & Error Detection:** Identifying transactions that deviate from expected patterns or violate basic business rules — such as negative quantities or prices, duplicate transaction IDs, invalid dates, or inconsistent category values — which signal data entry or system errors rather than trustworthy records.

Dataset Description

The project uses a retail transactions dataset (`dataset_ecommerce_transactions_data.csv`) containing 100,000 transactions across 8 columns, deliberately dirtied to simulate real-world data quality problems: roughly 5% missing Transaction_ID/Transaction_Date/Customer_ID/Quantity values, 33% missing or malformed Price values, and about 17% missing Transaction_Status.

Key Fields

- **Transaction_ID:** Unique transaction identifier; about 5% of rows are missing this value and ~6% contain duplicate IDs
- **Transaction_Date:** About 4.9% missing; roughly 63% of the remaining dates are invalid (e.g. 2025-02-30, 2023-13-01) and need repair or rejection
- **Customer_ID:** About 4.9% missing

- **Product_Name:** 5 categories (Tablet, Laptop, Smartphone, Headphones, Coffee Machine); ~5% of values are truncated fragments (e.g. "Lapt", "Ta", "T") from upstream data entry errors
- **Quantity:** About 5% missing; roughly 32% of non-missing values are negative, which is invalid for a physical quantity
- **Price:** Stored as text with currency symbols (e.g. "\$420.21"); ~33% missing or non-numeric, and ~30% of numeric values are negative
- **Payment_Method:** 7 raw variants of 3 real methods (Credit Card, PayPal, Cash) due to inconsistent casing, spacing, and abbreviation — e.g. "creditcard", "paypal", "PayPal "
- **Transaction_Status:** 5 raw variants of 4 real statuses (Completed, Pending, Failed) due to inconsistent casing — e.g. "complete", "Completed", "completed"; ~17% missing
- **Data Integrity Issues:** Combination of missing keys, duplicate Transaction_IDs, and invalid dates means a meaningful share of rows cannot be trusted without cleaning
- **Product_Name Truncation:** Beyond the 5 core products, dozens of truncated variants appear (e.g. "Coffee Ma", "Smartp", "L"), suggesting a source-system field length or encoding problem

System Architecture

Pipeline Architecture:

Transaction CSV → Data Ingestion Layer → Data Validation Engine → Data Cleaning & Standardization → Rule-Based Validation Engine → Statistical Outlier Detection → ML Anomaly Detection Engine → Quality Scoring Engine → AI Explanation Generator → Dashboard & Report Generator

Technology Stack

Layer	Technology / Tools	Purpose
Data Ingestion Layer	FastAPI / Express.js, Pandas	Upload and ingest CSV/Excel transaction datasets, validate file format, and load data into the system.
Data Validation Engine	Python, Pandas, NumPy, Regex (re)	Perform null checks, duplicate detection, datatype validation, range validation, regex validation, and business rule validation.

Data Cleaning & Standardization	Pandas, NumPy, Scikit-learn	Standardize inconsistent categorical text (payment method casing/spacing), strip currency symbols from Price, parse and repair malformed dates, handle missing values, and prepare a clean dataset.
Rule-Based Validation Engine	Python, Custom Rule Engine	Apply predefined data quality rules: flag negative quantities/prices, invalid or out-of-range dates, duplicate transaction IDs, and unrecognized payment-method or status values.
Statistical Outlier Detection	Python, Pandas, NumPy	Apply statistical methods (IQR, z-score) to flag extreme outliers in Price and Quantity beyond the rule-based thresholds.
ML Anomaly Detection Engine	Scikit-learn (Isolation Forest, Local Outlier Factor, DBSCAN)	Identify anomalous transactions using unsupervised machine learning algorithms to catch data quality issues not covered by explicit rules.
Quality Scoring Engine	Python, NumPy	Calculate an overall Dataset Quality Score, plus Field Completeness Score and Record Validity Score, based on validation results and anomaly detection output.
AI Explanation Generator	OpenAI GPT API / LLM API	Generate human-readable explanations of flagged data quality issues, assess business impact, and provide recommendations for cleaning or remediation.
Backend API	Node.js, Express.js	Manage APIs, coordinate data flow between frontend and ML modules, execute validation and anomaly detection pipelines, and return results.
Database	MongoDB	Store uploaded datasets, validation reports, anomaly logs, user information, and data-cleaning history.
Frontend Dashboard	React.js, Tailwind CSS, Axios / Streamlit	Provide an interactive user interface for dataset upload, anomaly visualization, quality metrics, and report downloads.
Data Visualization	Plotly, Chart.js	Create interactive charts, data quality trend analysis, anomaly distribution graphs, and executive dashboards.

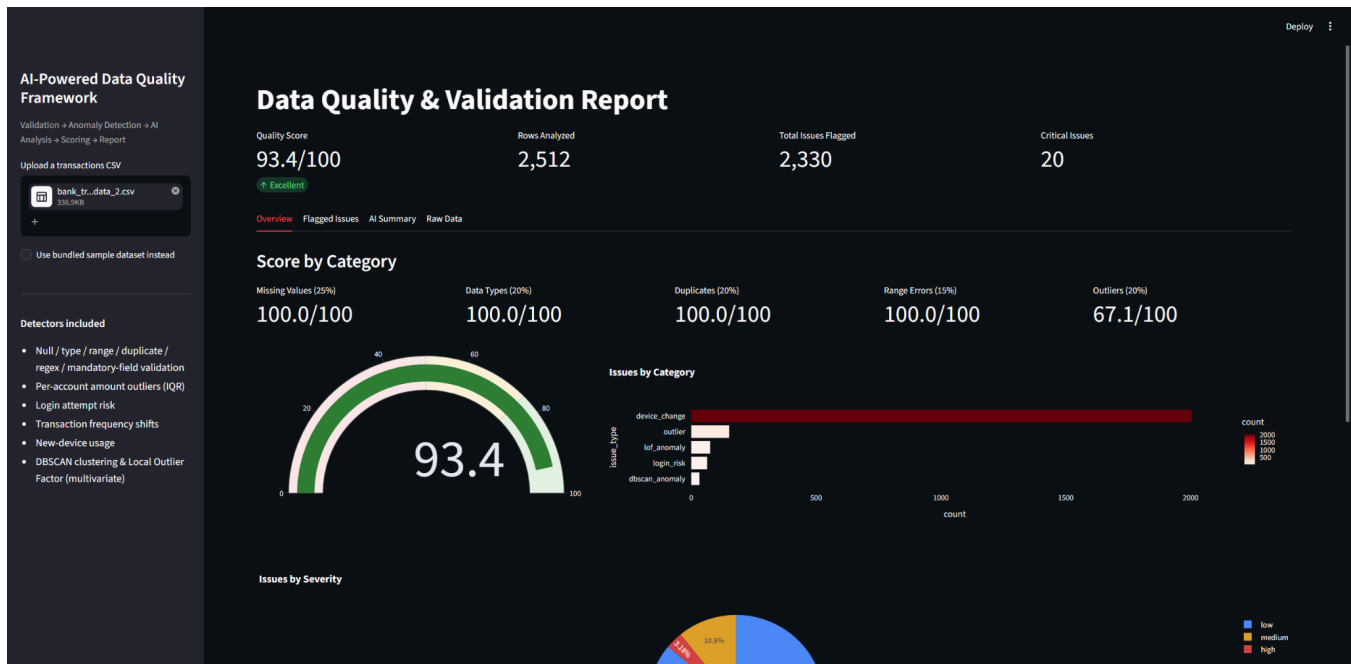
Report Generation	ReportLab, Python-docx, Python-pptx, Pandas	Generate downloadable PDF reports, PowerPoint summaries, Excel reports, and CSV exports for stakeholders.
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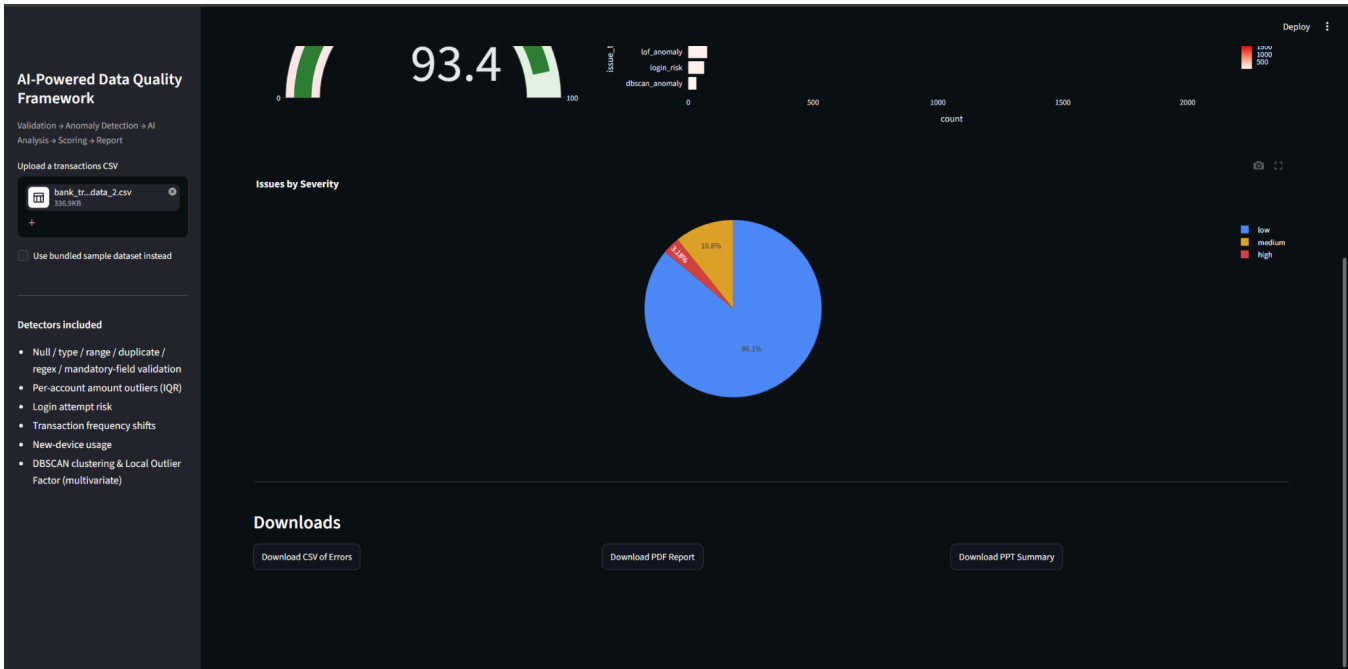
Sample Dashboard Design

This sample dashboard is built around a linear workflow — upload, validate, investigate, understand, export.

1. Upload / Landing Screen — Entry point for uploading a transaction dataset (CSV/Excel), with format validation on ingestion and a clear call-to-action to start analysis.
2. Overview Dashboard — Top-level summary once analysis completes: overall Dataset Quality Score, Field Completeness Score, and Record Validity Score displayed as headline metric cards, giving an at-a-glance read before drilling into detail.
3. Validation Results — A structured, filterable view of every validation issue (nulls, duplicates, type mismatches, range violations, format violations) with row-level detail, so analysts can trace exactly which records failed which rule.
4. Anomaly & Outlier Visualization — Interactive charts (Plotly/Chart.js) showing anomaly distribution, data quality trend analysis over time, and flagged transactions plotted against normal patterns — the visual layer for the ML Anomaly Detection Engine's output (Isolation Forest, LOF, DBSCAN).
5. Quality Scoring Breakdown — Detailed view of how the Quality Scoring Engine arrived at its scores, broken out by category/weight, so the score is explainable rather than a single opaque number.

6. AI Explanation Panel — Plain-English summaries generated by the AI Explanation Generator: what was flagged, why it matters (business impact), and recommended next steps, prioritized by severity.
7. Reports & Export — Download center for the PDF report, PowerPoint summary, Excel report, and CSV export of errors — the final deliverable layer for stakeholders who need the findings outside the dashboard.





AI-Powered Data Quality Framework

Validation → Anomaly Detection → AI Analysis → Scoring → Report

Upload a transactions CSV

Upload

200MB per file • CSV

☒ Use bundled sample dataset instead

Detectors included

- Null / type / range / duplicate / regex / mandatory-field validation
- Per-account amount outliers (IQR)
- Login attempt risk
- Transaction frequency shifts
- New-device usage
- DBSCAN clustering & Local Outlier Factor (multivariate)

Data Quality & Validation Report

Quality Score

93.4/100

↑ Excellent

Rows Analyzed

2,512

Total Issues Flagged

2,330

Critical Issues

20

Overview

Flagged Issues

AI Summary

Raw Data

All Flagged Issues

Filter by severity

high

medium

low

Filter by issue type

dbscan_anomaly

device_change

lof_anomaly

login_risk

outlier

row_id	column	issue_type	severity	message
0	DeviceID	device_change	low	Account AC00128 used a new device (D000380) not seen in its prior transactions
1	DeviceID	device_change	low	Account AC00453 used a new device (D000051) not seen in its prior transactions
2	DeviceID	device_change	low	Account AC00019 used a new device (D000235) not seen in its prior transactions
3	DeviceID	device_change	low	Account AC00070 used a new device (D000187) not seen in its prior transactions
4	DeviceID	device_change	low	Account AC00411 used a new device (D000308) not seen in its prior transactions
5	DeviceID	device_change	low	Account AC00393 used a new device (D000579) not seen in its prior transactions
6	DeviceID	device_change	low	Account AC00069 used a new device (D000500) not seen in its prior transactions
7	DeviceID	device_change	low	Account AC00264 used a new device (D000054) not seen in its prior transactions
8	DeviceID	device_change	low	Account AC00270 used a new device (D000466) not seen in its prior transactions
9	DeviceID	device_change	low	Account AC00317 used a new device (D000671) not seen in its prior transactions
10	DeviceID	device_change	low	Account AC00359 used a new device (D000432) not seen in its prior transactions

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HighMediumLow

Filter by issue type

dbscan_anomalydevice_changelof_anomalylogin_riskoutlier

	row_id	column	issue_type	severity	message	
	0	0	DeviceID	device_change	low	Account AC00128 used a new device (D000380) not seen in its prior transactions
	1	1	DeviceID	device_change	low	Account AC00455 used a new device (D000051) not seen in its prior transactions
	2	2	DeviceID	device_change	low	Account AC00019 used a new device (D000235) not seen in its prior transactions
	3	3	DeviceID	device_change	low	Account AC00070 used a new device (D000187) not seen in its prior transactions
	4	4	DeviceID	device_change	low	Account AC00411 used a new device (D000308) not seen in its prior transactions
	5	5	DeviceID	device_change	low	Account AC00393 used a new device (D000579) not seen in its prior transactions
	6	7	DeviceID	device_change	low	Account AC00069 used a new device (D000500) not seen in its prior transactions
	7	13	DeviceID	device_change	low	Account AC00264 used a new device (D000054) not seen in its prior transactions
	8	15	DeviceID	device_change	low	Account AC00270 used a new device (D000466) not seen in its prior transactions
	9	16	DeviceID	device_change	low	Account AC00317 used a new device (D000671) not seen in its prior transactions
	10	17	DeviceID	device_change	low	Account AC00359 used a new device (D000432) not seen in its prior transactions
	11	19	DeviceID	device_change	low	Account AC00285 used a new device (D000039) not seen in its prior transactions

Showing 2,330 of 2,330 total issues

Downloads

Download CSV of Errors

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Download PPT Summary

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OverviewFlagged IssuesAI SummaryRaw Data

AI-Generated Summary

This dataset of 2512 transactions scored 93.4/100 on data quality, with 2330 total issues flagged across 5 categories.

Score by category:

- Outliers: 67.1/100
- Missing Values: 100.0/100
- Duplicates: 100.0/100
- Data Types: 100.0/100
- Range Errors: 100.0/100

Key findings:

- 2005x device change: Some transactions came from a device not previously associated with the account.
- 152x outlier: Some transactions are unusually large or small compared to that specific account's normal spending pattern.
- 76x lof anomaly: Some transactions sit in a much sparser neighborhood than similar transactions — a density-based anomaly signal.
- 64x login risk: Several transactions were preceded by an unusually high number of login attempts, a common signal in fraud detection.
- 33x dbscan anomaly: Some transactions don't belong to any dense cluster of normal behavior across amount, age, duration, and login pattern.

Business impact (highest priority first):

- Outliers: These are the transactions most likely to represent real fraud or account compromise — this is where investigative time should go first.

Recommended next steps:

- Cross-reference with location and IP address; a single new device is common, but paired with other flags it's higher risk.

AI-Powered Data Quality Framework

Validation → Anomaly Detection → AI Analysis → Scoring → Report

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Outliers: 67.1/100

Missing Values: 100.0/100

Duplicates: 100.0/100

Data Types: 100.0/100

Range Errors: 100.0/100

Key findings:

- 2005x device change: Some transactions came from a device not previously associated with the account.
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Business impact (highest priority first):

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Recommended next steps:

- Cross-reference with location and IP address; a single new device is common, but paired with other flags it's higher risk.
- Route these accounts to manual review; consider a secondary check against merchant category and location.
- Review manually; this transaction sits in an unusually sparse neighborhood relative to similar transactions, a common fraud signal.

Using rule-based fallback summary (no ANTHROPIC_API_KEY set). Set the environment variable to enable live LLM-generated narratives.

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Overview

Flagged Issues

AI Summary

Raw Data

Raw Uploaded Data

	TransactionID	AccountID	TransactionAmount	TransactionDate	TransactionType	Location	DeviceID	IP Address	MerchantID	Channel	CustomerAge	CustomerOccupation	TransactionDuration	LoginAttempts	AccountRisk
0	TX000001	AC00128	14.09	2023-04-11 16:29:14	Debit	San Diego	D000380	162.196.218.92	M015	ATM	70	Doctor	81	1	511
1	TX000002	AC00435	376.24	2023-06-27 16:44:39	Debit	Houston	D000651	13.149.61.4	M052	ATM	68	Doctor	141	1	1375
2	TX000003	AC00019	126.29	2023-07-10 18:16:08	Debit	Mesa	D000235	215.97.143.157	M009	Online	19	Student	56	1	112
3	TX000004	AC00070	184.5	2023-05-05 16:32:11	Debit	Raleigh	D000187	200.13.225.150	M002	Online	26	Student	25	1	856
4	TX000005	AC00411	13.45	2023-10-16 17:51:24	Credit	Atlanta	D000308	65.164.3.100	M091	Online	26	Student	198	1	74
5	TX000006	AC00393	92.15	2023-04-03 17:15:01	Debit	Oklahoma City	D000579	117.67.192.211	M054	ATM	18	Student	172	1	76
6	TX000007	AC00199	7.08	2023-02-15 16:36:48	Credit	Seattle	D000241	140.212.253.222	M019	ATM	37	Doctor	139	1	1331
7	TX000008	AC00069	171.42	2023-05-08 17:47:59	Credit	Indianapolis	D000500	92.214.76.157	M020	Branch	67	Retired	291	1	275
8	TX000009	AC00135	106.23	2023-03-21 16:59:46	Credit	Detroit	D000690	24.148.92.177	M035	Branch	51	Engineer	86	1	905
9	TX000010	AC00385	815.96	2023-03-31 16:06:57	Debit	Nashville	D000199	32.169.88.41	M007	ATM	55	Doctor	120	1	102
10	TX000011	AC00150	17.78	2023-03-14 16:46:10	Credit	Albuquerque	D000205	213.15.9.253	M073	Online	52	Engineer	59	1	755
11	TX000012	AC00459	190.02	2023-02-06 17:30:00	Debit	Memphis	D000589	116.175.11.222	M030	Online	21	Student	173	1	152

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Project Structure

